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European Patent Application No. 22727010.5 (4330677)
Regional Phase of PCT Application No. PCT/EP2022/061385
Elypta AB

I write in response to the Communication pursuant to Article 94(3) EPC which is dated 9 April 2025, the current deadline for responding to which is 9 October 2025.

Claim Amendments

We enclose herewith an amended claim set upon which further prosecution of this application should be based. A "marked-up" version of the enclosed amended set showing the amendments made as "tracked changes" as compared to the claims currently on file (i.e. as compared to the claims filed on 16 May 2024) is also enclosed for ease of reference.

In the enclosed amended claim set, claim 1 has been amended to recite that the sample:

"… has been subjected to processing prior to determining said level and/or chemical composition,
wherein said processing
(a) comprises fragmenting said one or both GAGs into disaccharide units; and
(b) does not comprise prior to (a):
(i) contacting said sample with a proteolytic agent; or
(ii) purifying said one or both GAGs in said sample based on the negative charge of said GAGs."

Basis for this amendment is found, for example, in PCT claims 21 and 22. Claim 15 has been amended for consistency with amended claim 1.

Previous claims 10 and 11 have been removed.

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Preferred and optional features that were recited in part (a) of previous claim 11 are now recited in new dependent claims 10 and 11. Basis for new dependent claims 10 and 11 is provided, for example, by PCT claims 23 and 24.

The amendments have been made in the interests of expediting prosecution and without prejudice. No subject-matter has been abandoned as a result of the amendments filed herewith. The Applicant reserves the right to re-introduce subject-matter or to file one or more divisional applications to any appropriate subject-matter.

Response to Objections in the Communication pursuant to Article 94(3) EPC

In ***item 1.1*** of the Communication pursuant to Article 94(3) EPC, the Examiner has indicated that an expression in previous claim 10 does not comply with Article 123(2) EPC and also lacks clarity. As described above, previous claim 10 has now been removed, thus rendering this objection moot.

In ***item 2*** of the Communication pursuant to Article 94(3) EPC, the Examiner has referred to the IPRP and indicated that the objections raised in the IPRP give rise to objections under the corresponding provisions of the EPC.

For the reasons given in ***item 1.1.1*** of the IPRP, the Examiner has alleged that previous claim 1 lacks novelty over D1. The Examiner states that the method of D1 "*can comprise treatment with i.a. in reality proteinase K or anion exchange, but not necessarily*". However, there is no direct and unambiguous disclosure in D1 of a method which does **not** comprise a step of contacting the sample with a proteolytic agent in accordance with amended claim 1 filed herewith, let alone a direct and unambiguous disclosure in D1 of a method which (i) does **not** comprise a step of contacting the sample with a proteolytic agent in accordance with amended claim 1 **and** (ii) does **not** comprise a step of purifying GAGs in the sample based on the negative charge of GAGs in accordance with amended claim 1. Thus, we respectfully submit that amended claim 1 (and all of the other claims) is novel over D1.

For the reasons given in ***item 1.1.2*** of the IPRP, the Examiner has alleged that previous claim 1 lacks novelty over D2. Of particular note, the Examiner states that the D2 method "*does not comprise negative charge purification*". We respectfully disagree with the Examiner's objection, as D2's method explicitly does include a negative charge purification step through the use of Cetavlon to precipitate sulfated GAGs from urine. In this regard, as part of its "Extraction and Identification of Glycosaminoglycans" section (D2, page 116), D2 explicitly details a purification step: "*The pH of the urine was adjusted to 6.0 with HCl, and 17 mL of a 5% solution of Cetavlon per liter of urine was added. After standing overnight in the cold, the precipitate formed was collected by centrifugation...*". Cetavlon (cetyltrimethylammonium bromide) is a cationic detergent commonly used to precipitate and thus purify sulfated GAGs based on their negative charge. D2 does not disclose a method that does **not** comprise a step of purifying GAGs in the sample based on the negative charge of GAGs in accordance with amended claim 1. For at least this reason, we respectfully submit that claim 1 (and all of the other claims) is novel over D2.

For the reasons given in ***item 1.1.3*** of the IPRP, the Examiner has alleged that previous claim 1 lacks novelty over D3. However, in section 2.2 of D3 titled "Protein Release" (D3, page 3), D3 states: "*An amount of 50 µL pronase E (20 mg/mL) ... was added to 50 µL serum aliquots. ... to achieve protein digestion*." D3 further states that "*the protein moiety of proteoglycans was first cleaved using pronase, a cocktail of proteases*" (D3, page 4, second sentence of section 3). Pronase is a proteolytic enzyme. D3 does not disclose a method that does **not** comprise a step of contacting the sample with a proteolytic agent in accordance with amended claim 1. For at least this reason, we respectfully submit that claim 1 (and all of the other claims) is novel over D3.

We would additionally point out to the Examiner that although the International Search Report indicates that the D3 was published on 1 January 2021, it seems that D3 was *actually* published on 23 June 2021 (see about halfway down the left-hand margin on the first page of D3), which is after the earliest priority date of the present application.

For the reasons given in ***item 1.1.4*** of the IPRP, the Examiner alleged that previous claim 1 lacks novelty over D4. We note that the D4 method does not involve a step of fragmenting GAGs into disaccharide units as required by amended claim 1. Indeed, we note that PCT claims 21 and 22 were not included in the D4-based novelty objection in the IPRP. As described above, claim 1 has been amended to recite the features of PCT claims 21 and 22. For at least the above reasons, we submit that amended claim 1 (and all of the other claims) are novel over D4.

We further submit that there is no suggestion of a method as presently claimed in any of the cited prior art documents. None of the cited prior art documents motivate the skilled person to develop a method which comprises fragmenting GAGs into disaccharide units (as per part (a) of amended claim 1), but which does **not** comprise prior to said fragmenting a step of contacting the sample with a proteolytic agent (as per part ((b)(i) of amended claim 1) **and** does **not** comprise prior to said fragmenting a step of purifying GAGs in the sample based on the negative charge of the GAGs (as per step (b)(ii) of claim 1). The claimed method advantageously gives rise to improvements and efficiencies in the GAG processing steps required for GAG-based cancer screening methods, for example in simplifying sample preparation.

For the reasons given in ***item 2.1*** of the IPRP, the Examiner alleged that the term “protein-free fraction” in claim 1 is unclear. We respectfully submit that the amendment to claim 1 (described above) overcomes this objection.

In view of the amendment to claim 1, and removal of dependent claim 22, we submit that the clarity objection in ***item 2.2*** of the IPRP has been rendered moot.

Concluding remarks

It is hoped that in view of the amendments filed herewith, the application may be viewed as in order for allowance. As a precaution, however, should any objections remain, appointment of Oral

Proceedings is requested. However, the Examiner is requested to contact me by telephone or in writing concerning any outstanding objections before any such proceedings are appointed.

Yours faithfully
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